

Secure Exit

SIMATS
ENGINEERING

SIMATS Online Programming Lab

JAVA PROGRAMMING

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CO3-AT1-Assignment

View Only

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QUESTIONS

Q1

Banking Transaction System –
Exception Handling
10 marks

Q2

Hospital Patient Registration –
User-Defined Exception
10 marks

Q3

E-Commerce Order Processing
– Built-in Exceptions
10 marks

Q4

Online Food Delivery – Multiple
Threads and Execution Order
10 marks

Q5

Manufacturing Plant –
Producer-Consumer Problem
10 marks

5/5 answered

Q1

Banking Transaction System –
Exception Handling10
marks

Develop a Java program for an ATM withdrawal system. The program accepts the account balance and withdrawal amount.

The program must:

- Accept the current account balance and withdrawal amount.
- Throw a user-defined `InsufficientBalanceException` when the withdrawal amount exceeds the balance.
- Handle withdrawal amounts that are zero or negative.
- Handle invalid numeric input using a suitable built-in exception.
- Display a meaningful transaction result.
- Use try-catch-finally so that the transaction completion message is always displayed.

Sample Test Cases:

Input	Scenario	Expected Result	Purpose
10000	3000	Normal case	
10000	12000	Boundary / exception case	
10000	-500	Invalid input	
abc	3000	Invalid input	

Your Code

```
1 import java.util.*;
2 class ATM{
3     public static void main(String[] args)
4     {
5         Scanner sc = new Scanner(System.in);
6         try{
7             double balance = sc.nextDouble();
8             double amount = sc.nextDouble();
9             if (amount <= 0)
10                 throw new IllegalArgumentException("Amount should be positive");
11             if (amount > balance)
12                 throw new InsufficientBalanceException("Insufficient balance");
13             System.out.printf("Transaction Successful. Remaining balance: %.6f\n", balance - amount);
14         } catch (InsufficientBalanceException e) {
15             System.out.println(e.getMessage());
16         } catch (IllegalArgumentException e) {
17             System.out.println(e.getMessage());
18         } catch (InputMismatchException e) {
19             System.out.println(e.getMessage());
20         } finally{
21             System.out.println("Transaction completed");
22         }
23     }
24 }
25
26 class InsufficientBalanceException extends RuntimeException{
27     InsufficientBalanceException(String msg){
28         super(msg);
29     }
30 }
```



Input

10000
3000

Output

Transaction Successful.Remaining balance:
7000.000000
Transaction completed

✔ Submitted

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